

# From neighbour counts to source-aware inference: a calibration audit of spatial perturbation screens

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Spatial perturbation transcriptomics can reveal how genetically edited cells affect surrounding tissue, but the neighbouring cells around one source are not independent perturbation replicates. We developed Perturbation PSF, a source-aware calibration audit for radial perturbation-associated contrasts, and applied it to a mouse-brain Spatial Perturb-seq dataset containing 229,775 segmented cells, 17 disease-gene perturbations, and a safe-harbour control. On the identical biological comparison, treating 24,453–31,189 source–recipient pairs as independent produced 16 of 20 nominal program-level hits. Recomputing uncertainty at the 78 or 101 perturbation and control sources inflated standard errors by a median of 5.5-fold and eliminated every hit (0/20). Restricted wild cluster bootstrap, source-label permutation, equal-source weighting, alternative adjustment sets, and leave-one-mouse-out analyses agreed. Two near-field *Lrrk2* contrasts fell in the extreme tail of a matched pseudo-source diagnostic, but were method-dependent and unsupported by the primary global test; we report them as leads, not findings. The analysis exposes recipient reuse, interference, animal imbalance, and source-level power as central design constraints.  $K(r)$  is a cross-sectional adjusted contrast, not an identified causal impulse response.

spatial transcriptomics | Perturb-seq | clustered inference | interference | point-spread function

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## Introduction

Point-spread functions (PSFs) turn a localized input into a spatial description of a system. In optics, their width quantifies resolution; in linear systems they are closely related to Green's functions. We previously applied this perspective to multi-resolution hash encodings for three-dimensional neural reconstruction, using a spatial kernel to quantify resolution, grid anisotropy, and collision noise (1). That work motivated a broader question: can physical-systems thinking make spatially distributed biological measurements more interpretable? Perturb-seq couples pooled genetic perturbations to single-cell transcriptomes (2). Spatial Perturb-seq combines this logic with Stereo-seq's large field of view and near-cellular spatial resolution (3, 4), preserving the distance between an edited source and surrounding barcode-negative recipients. A radial response profile could distinguish a contact-scale effect from a weaker long-range effect and express both magnitude and reach in physical units.

This question sits within a rapidly growing field. Perturb-FISH

directly measures intrinsic and intercellular consequences of pooled CRISPR perturbations in spatial context (5). CONCERT learns perturbation-specific Gaussian-process spatial kernels for niche-aware prediction on extended perturbation regions (6). BANKSY embeds local molecular neighbourhoods for spatial domain discovery (7), whereas COMMOT models spatial ligand–receptor communication via collective optimal transport (8). Our contribution is not the first spatial perturbation kernel. It is an inferential audit for sparse, single-cell sources: how much apparent evidence changes when the perturbation source, rather than every neighbouring recipient, defines the replication unit.

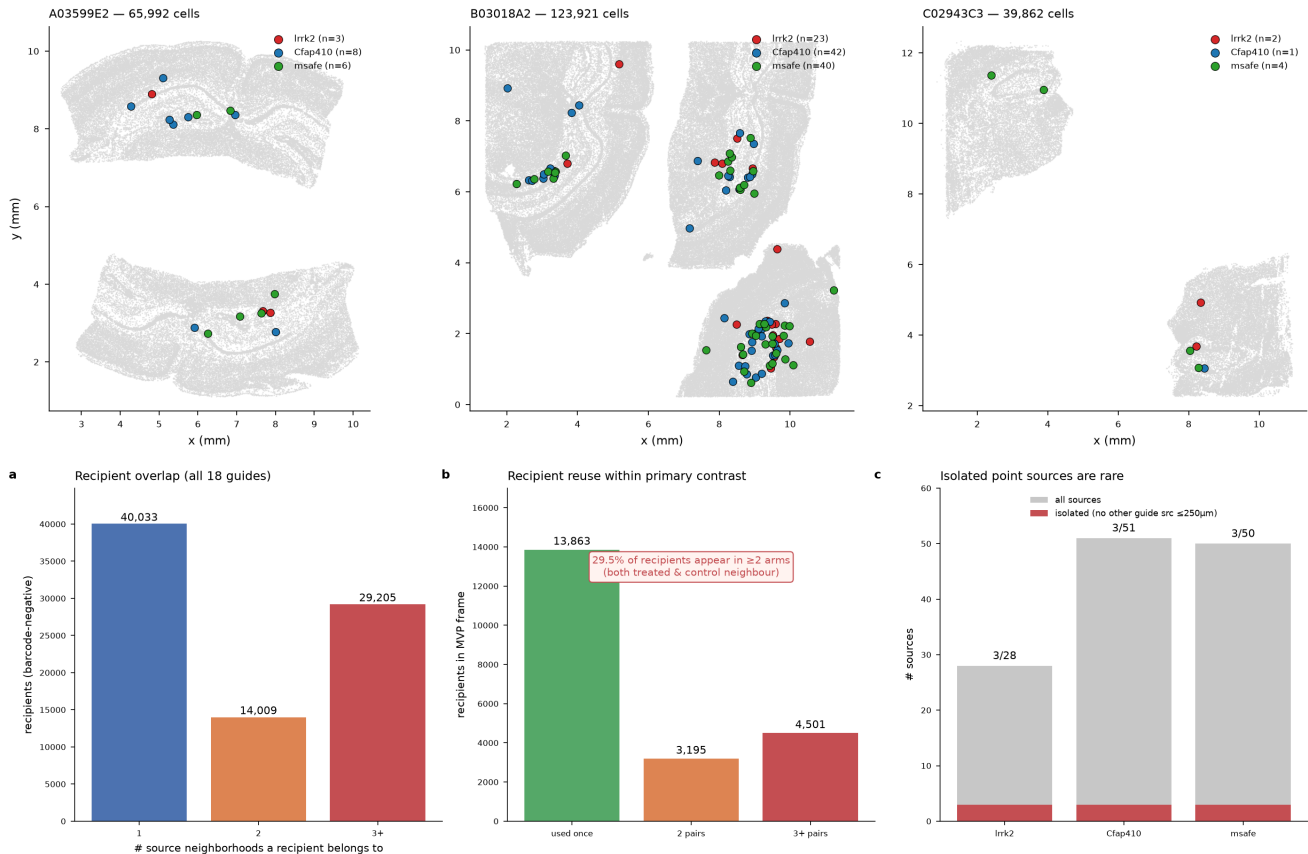
The PSF analogy must also be limited. Perturbation time is unknown, edit success is inferred from barcode detection, tissue is heterogeneous, and several sources may influence the same recipient. We therefore estimate an adjusted radial association  $K(r)$  rather than a causal impulse response. The project began by searching for a biological PSF; statistical review redirected it toward the more general question of whether the available experiment could resolve one.

## Results

**Sparse sources share a reused recipient field.** We analyzed three cell-segmented Stereo-seq sections from GSE274447. Coordinates were converted using the GEF root metadata (`resolution=500 nm`), and 222,140 of 229,775 segmented cells passed quality control. Strict single-guide calling at  $\geq 2$  guide UMIs yielded 28 *Lrrk2*, 51 *Cfap410*, and 50 *msafe* control sources for the primary comparisons.

The sources are sparse, but 250  $\mu\text{m}$  neighbourhoods overlap strongly (Figure 1). Across all guides, 37.8% of barcode-negative cells lie near at least one source, and 50.6% of exposed recipients neighbour at least two distinct perturbations. The primary three-guide frame contains 40,087 source–recipient pairs but only 21,559 unique recipients; 29.5% of recipients appear in at least two treatment arms. This geometry creates within-source correlation and treatment interference, violating the no-interference component of SUTVA (9).

**Pair-level significance disappears under source-aware uncertainty.** For each outcome we fitted a cubic regression spline in distance, a treatment-by-distance interaction, recipient cell type, spatial niche, section, depth, source class, and pre-specified source-level covariates. The primary scalar



**Fig. 1. Sparse sources and shared recipients.** Source cells are sparse across three sections, but their recipient neighbourhoods overlap. Recipient reuse motivates source-aware uncertainty and explicit interference diagnostics.

statistic averaged the adjusted contrast within 40  $\mu\text{m}$ . The testing family comprised ten programs for each of two perturbations (20 contrasts), with Benjamini–Hochberg correction applied within this pre-specified family (10).

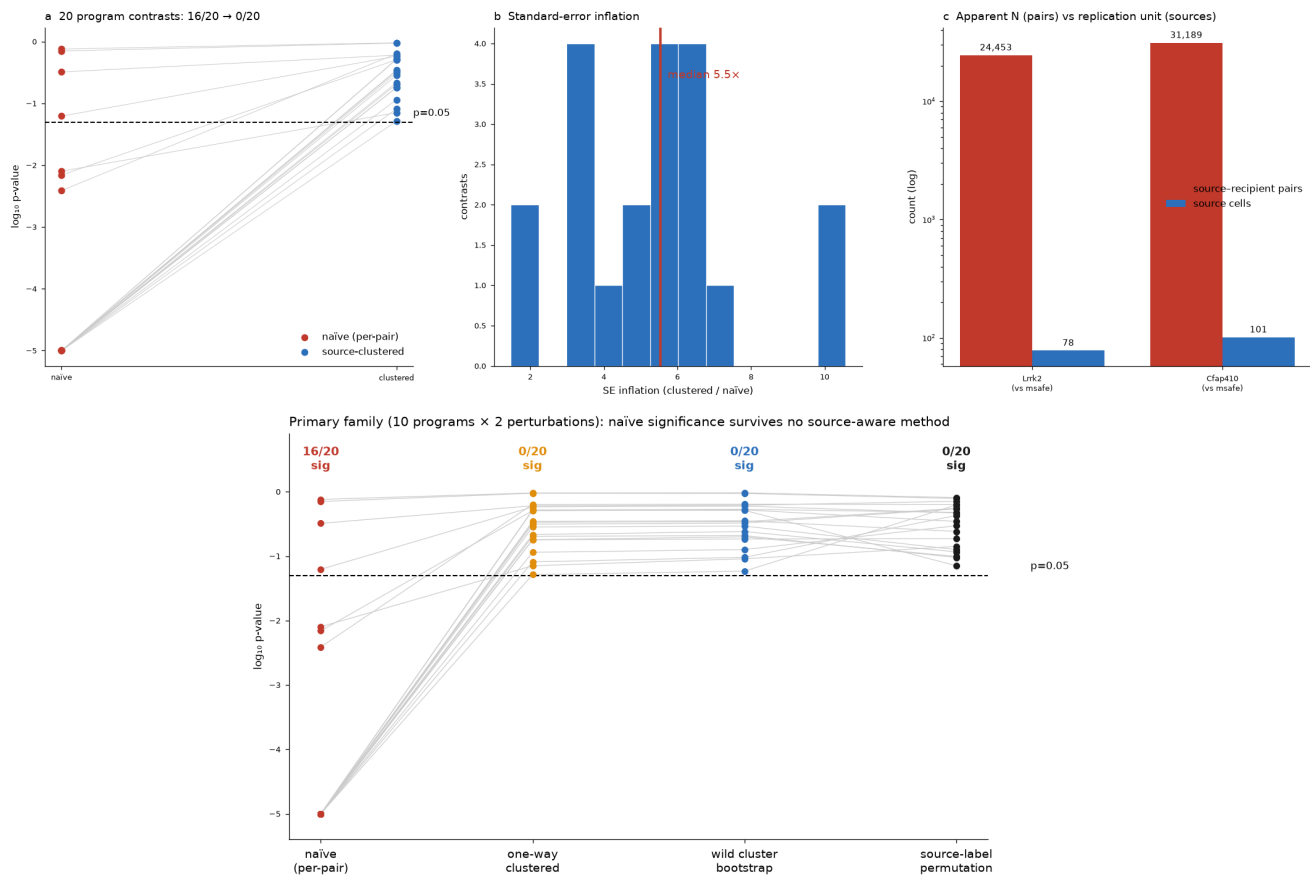
The central ablation holds the comparison and point estimates fixed while changing the covariance assumption. Naive pair-level inference reported 16/20 nominal hits. Source-clustered uncertainty, which permits arbitrary residual correlation within source (11), increased standard errors by a median of 5.5-fold (maximum 10.6-fold), leaving 0/20 significant (Figure 2). Because ordinary cluster-robust asymptotics can be unreliable with few clusters, we used a restricted wild cluster bootstrap (12); it also returned 0/20. A stratified source-label permutation and a Freedman–Lane residual permutation (13) likewise produced 0/20 after correction. The result does not imply that every biological effect is exactly zero. It shows that the available sources do not justify the precision implied by counting tens of thousands of pairs as independent experiments.

**The verdict survives estimand and animal sensitivity analyses.** Pair weighting gives more influence to sources with more recipients. We therefore residualized the baseline model, reduced every source to one near-field summary, and compared arms with equal source weights. Mann–Whitney and inverse-cluster-size OLS again produced 0/20 FDR-supported

contrasts. Three adjustment sets—a total local effect without composition or niche terms, the primary within-type specification, and a no-niche specification—each returned 0/20 (Figure 3). Thus the result is not created by recipient-count weighting or by conditioning on a possible post-treatment mediator.

The deposited samples map three chips one-to-one to three mice, while the article text reports four mice (4). We use the deposited mapping and treat mouse/chip as the highest blocking level. Mouse 2 contains 23/28 *Lrrk2* and 42/51 *Cfap410* sources: 82% of each perturbation. Leave-one-mouse-out analysis gives 0/20 FDR-supported contrasts under every drop. Inference is conditional on these realized tissues and cannot support population-level generalization.

**Two *Lrrk2* contrasts remain exploratory leads.** A matched pseudo-source diagnostic sampled two pseudo-arms matched on section, niche, cell type, depth, density, and boundary distance. Two near-field *Lrrk2* contrasts—endolysosomal and astrocyte-reactivity programs—fell in the extreme tail ( $q \approx 0.05$ – $0.09$ ). A non-overlap nearest-source design, in which each recipient is assigned to at most one source, identified the same two nominal contrasts but 0/20 after correction. We do not promote them to findings. They are unsupported by the primary global profile test, arise from one *Lrrk2*-targeting construct, and are concentrated in one mouse. They also



**Fig. 2. Replication-unit ablation.** The same contrasts appear highly significant when pairs are independent. Source-aware uncertainty eliminates all 20 primary hits; wild bootstrap and source-label permutation confirm that this is not a peculiarity of one covariance estimator.

weaken under a cross-cell quantile-normalized rank average. Rank-based single-cell signature methods such as UCell instead compute a within-cell Mann–Whitney statistic (14); we explicitly do not label our sensitivity score as UCell. Technical exclusion tests show that the candidates survive removal of touching or sub-soma-diameter pairs ( $r_{\min} \geq 20 \mu\text{m}$ ), reducing concern about segmentation leakage without overcoming the biological replication limits (Figure 4).

**Source-level power is the experimental bottleneck.** Idealized source-level simulations injected exponential near-field signals into the observed geometry. At the observed allocations, the estimated 80% minimum detectable effect is approximately 0.17–0.19 for *Lrrk2* (28 versus 50 sources) and 0.14–0.16 for *Cfap410* (51 versus 50), overlapping plausible observed effects. Balanced designs require on the order of  $10^2$  sources per arm for effects of this scale. These are optimistic lower bounds because the simulation omits count noise, dropout, guide-call uncertainty, annotation error, and animal heterogeneity. Real studies require more sources distributed across more animals (Figure 5).

## Methods

**Cell annotations, niches, and program scores.** Genes were aligned across sections, counts normalized to  $10^4$  per cell and log transformed, and 2,000 highly variable genes

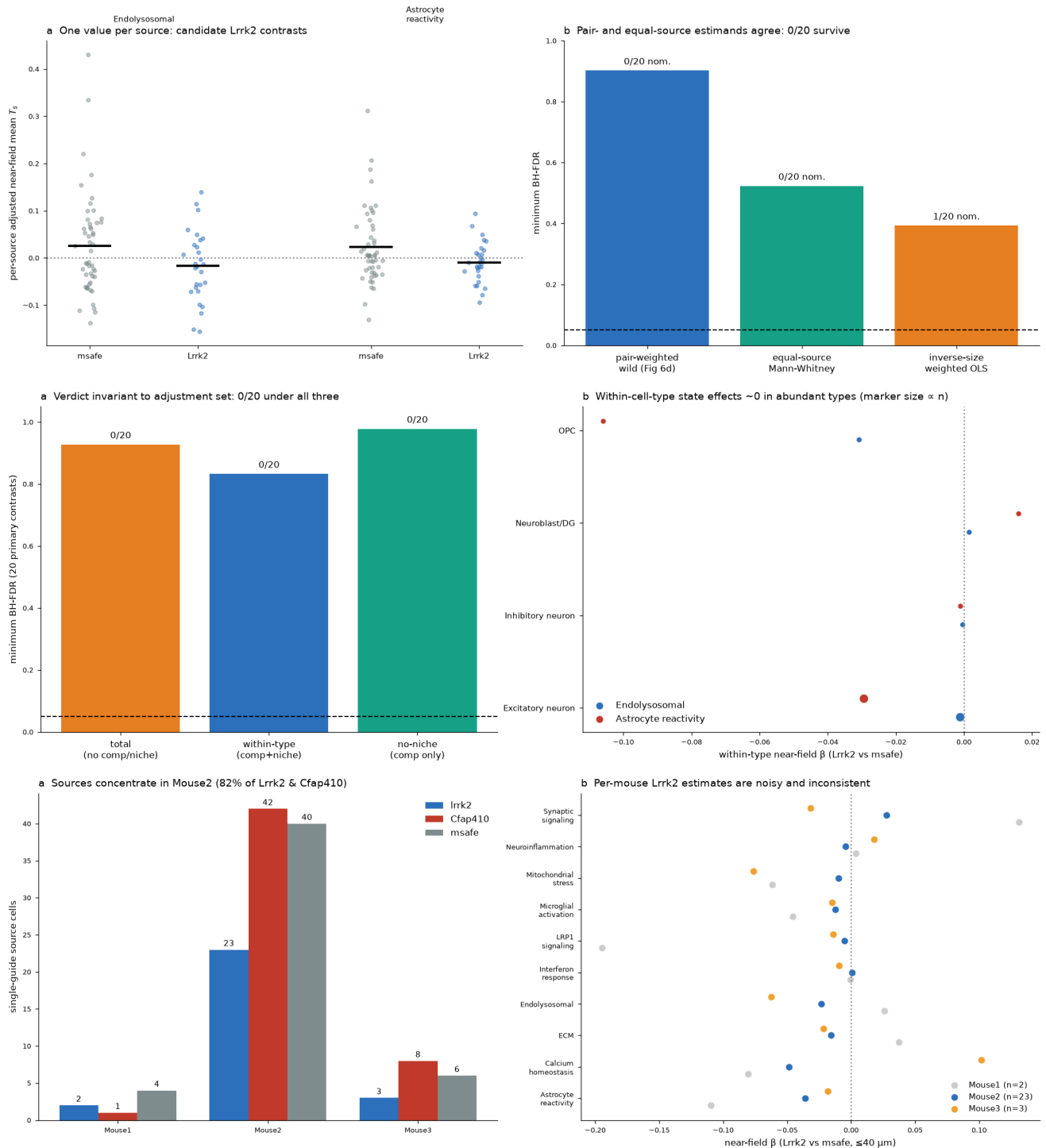
reduced to 30 principal components. Relative enrichment against brain marker panels produced ten broad cell classes. Spatial niches were obtained with a BANKSY-like augmentation of each cell’s embedding by the mean embedding of its 16 nearest spatial neighbours (7). Primary program scores used the control-gene-subtracted approach popularized for single-cell state signatures (15); a separate cross-cell rank sensitivity tested score dependence.

**Radial contrast and diagnostics.** For outcome  $Y$ , recipient  $i$ , source  $s$ , distance  $r$ , and source treatment  $T_s$ , we fitted

$$Y_{is} = \alpha + f(r_{is}) + T_s\beta + T_sg(r_{is}) + \mathbf{x}_{is}^T\boldsymbol{\gamma} + \varepsilon_{is}, \quad (1)$$

where  $f$  and  $g$  are cubic regression splines with knots at 30, 60, 100, and 160  $\mu\text{m}$ . The radial contrast is  $K(r) = \beta + g(r)$ . Our primary statistic averages  $K(r)$  over  $r \leq 40 \mu\text{m}$ . Diagnostics include wild cluster bootstrap, stratified label and residual permutation, equal-source summaries, matched pseudo-sources, nearest-source non-overlap, alternative adjustment sets, leave-one-mouse-out, sub-diameter exclusion, and idealized power recovery.

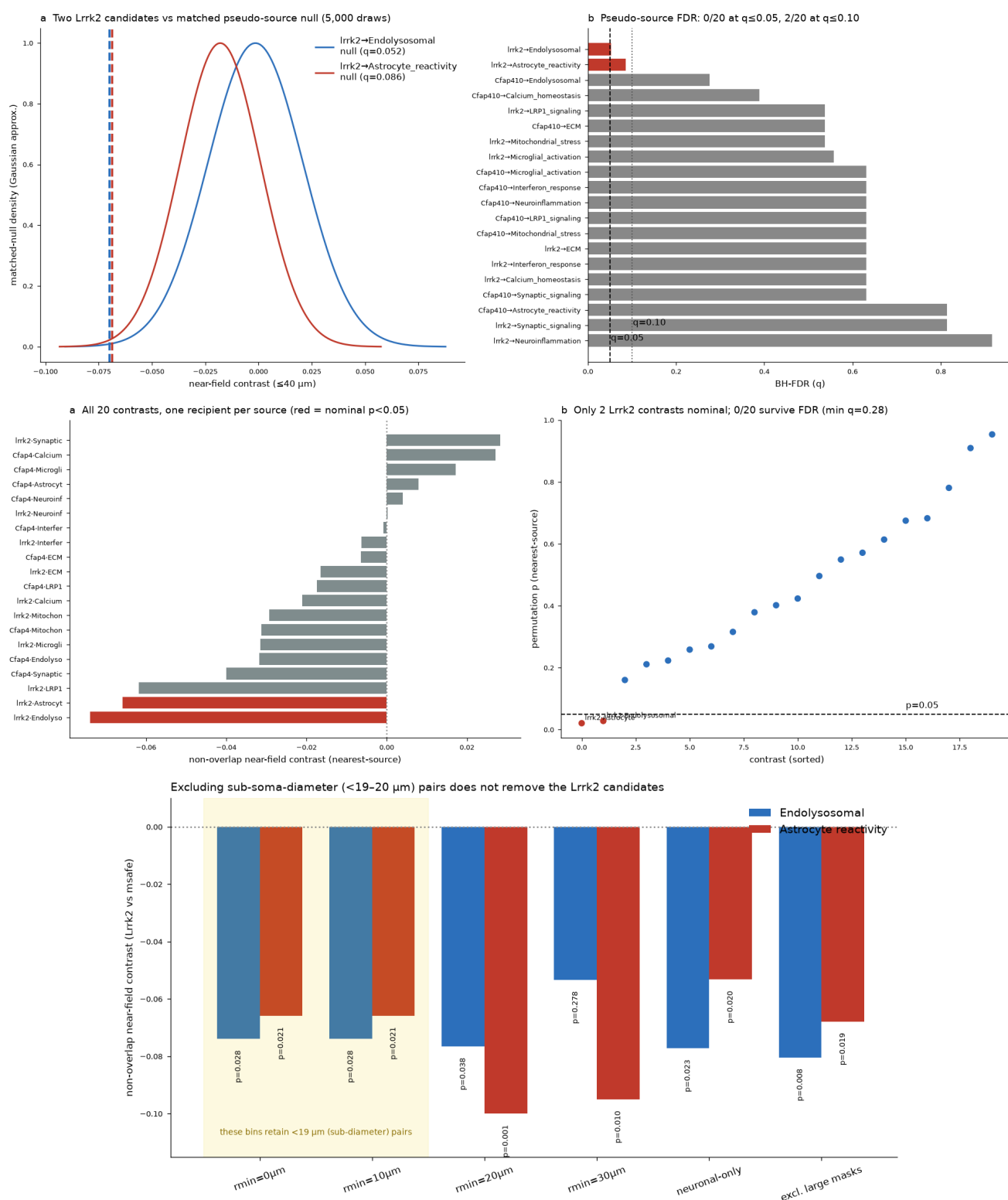
**Human-directed Claude Science iteration.** The review unfolded as a human-directed conversation with the evidence. After each result, the author inspected what had changed, identified the next statistical or design concern, and redirected the



**Fig. 3. Robustness of the source-aware verdict.** Equal-source and pair-weighted estimands agree; total-effect, within-type, and no-niche adjustment sets all yield 0/20; and 82% of both perturbations' sources lie in one mouse.

analysis. An apparently strong screen first prompted an audit of the replication unit; the resulting collapse prompted small-cluster and overlap stress tests; inspection of those tables exposed a testing-family bookkeeping problem; and the corrected output led back to mouse balance, estimand choice, module-score dependence, and cell-diameter leakage. Claude Science translated each evolving concern into literature checks, GEF inspection, executable analyses, di-

agnostics, reruns, and rebuilt artifacts while maintaining the full research state. The critical judgments and direction were human; Claude Science made each next question testable at scale. The decision log and iteration record are included in the repository.



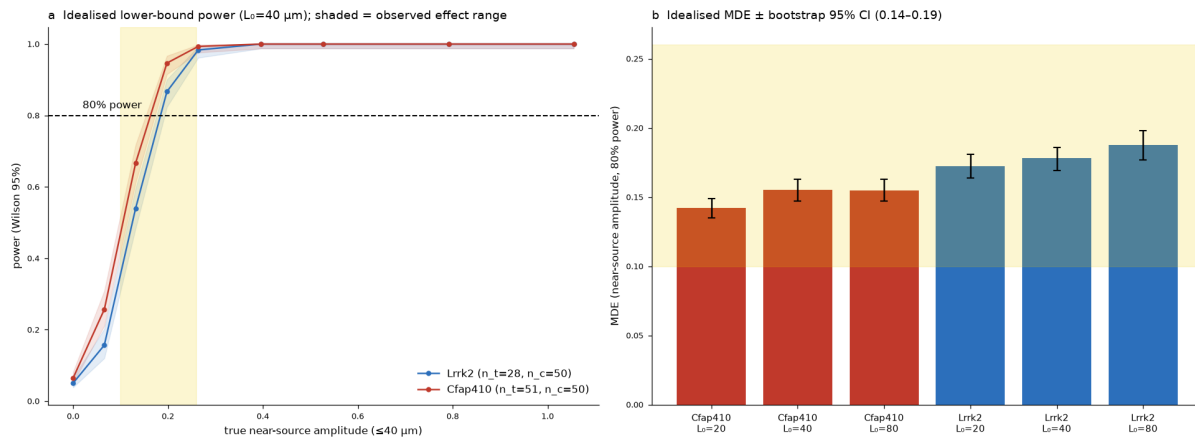
**Fig. 4. Candidate audit.** Matched pseudo-sources and a non-overlap design nominate the same two *Lrrk2* contrasts, but none of the primary family survives FDR correction. Excluding sub-soma-diameter pairs does not remove the candidates.

## Discussion

The strongest result is statistical: pair-level inference can make a sparse spatial perturbation screen appear far more precise than its intervention units justify. In this dataset, respecting the source changes 16/20 nominal program hits to 0/20, with

resampling, permutation, equal-source, and mouse sensitivity analyses agreeing. This dependence structure is not unique to one assay; it arises whenever many spatial outcomes are collected around a small number of perturbed sources.

The work also illustrates what a PSF perspective contributes



**Fig. 5. Observed-allocation power analysis.** Bootstrap intervals for logistic-fit minimum detectable effects. The calculation is a lower-bound design aid, not a guarantee of discovery at a particular source count.

and where it stops. Distance-resolved profiles separate magnitude from reach, but the observed  $K(r)$  is not a Green's function. Cross-sectional association, recipient reuse, multiple nearby sources, and animal imbalance prevent a causal PSF interpretation. Future experiments should use multiple constructs per gene, prospectively balance many more sources across animals, and model each recipient's joint exposure to all nearby sources. Two-way source–recipient dependence and explicit interference estimands are natural next steps. The original question was whether cell activation has a PSF-like correlation. The defensible answer is a calibrated boundary: suggestive local structure exists, but the present experiment does not resolve a population-level response. The audit converts that boundary into actionable design requirements for the next spatial perturbation screen.

## Data and code availability

Raw data are available from GEO accession GSE274447; the dataset authors provide code and an archived release (4). The submission repository contains LaTeX source, the source-aware audit package, review-validated tables, figures, tests, and demo assets. The 2 GB raw archive and large intermediate matrices are regenerated rather than vendored.

## ACKNOWLEDGEMENTS

This work was built during the 2026 Built with Claude: Life Sciences hackathon. The author directed the statistical review by inspecting each result and deciding which assumption to challenge next. Claude Science served as the executable research environment that investigated those concerns, traced issues, reran analyses, and rebuilt the paper and figures. This back-and-forth materially changed the estimand, inference, bookkeeping, and strength of the final claims.

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